PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors)

Heat and material balance, line sizing and hydraulics

PVC-V1-PE-HMB-001

Heat and material balance of R-101 for the reference batch (5.27 h reaction, 43.2 t PVC per batch, peak heat release 3.43 MW at X 0.72) with two off-design batches (CW 38 °C, initiator × 1.3); batch heat balance and utilities (1708 m³ cooling water and 3.17 t LP steam per batch); line list of 16 lines with sizes; P-101A/B and P-103 hydraulics; design basis and margins; 67 ✓ / 3 ✗ checks against the model and the other documents.

Unit	PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors)
Prepared for	Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University
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0	2026-10-27	Issued for teaching	Apinan Soottitantawat		

The numbers in this document come from the PVC Suspension Reactor Ver 1 simulator model (R-101) and the documents of the same OTS, or are labelled "assumed" or "typical". It was generated for teaching: it is not a design deliverable and must not be used for construction or purchase. Checked and Approved are left blank on purpose.

1 Scope and basis

This document gives the heat and material balance (HMB) of reactor R-101, one of four identical 140 m³ batch suspension-PVC reactors, with two off-design batches, the heat balance of the batch and the utilities, the line list with sizes, the hydraulics of the jacket circulation pumps P-101A/B and the VCM charge pump P-103, and the design basis and margins. All numbers come from the operator-training-simulator model (the reference batch, the design state 2.0 h into the reaction, or a model run) or are labelled assumed or typical.

Basis. Recipe 60 t water + PVA and 51 t VCM, initiator dose 1, TIC-100 57 °C, CW supply 32 °C (model K). Reaction step = the reference batch pvBatch({}) (5.27 h, X 0.848, 43.2 t PVC, K 66.99). Charge steps are a scripted model run from an empty reactor (procedure order of the Transition tab); short-stop, degassing and discharge continue the reference batch on the same model copy. Design state = snapDesign (2.0 h into the reaction). Precomputed 2026-10-27 by tabs/eproc/tools/compute.js and re-checked in part 7.

1.1 What process engineering delivers, and what is different on a batch reactor

Order of the deliverables: PFD-100 → HMB → line list and line sizing → hydraulics (pump heads, NPSH, control-valve pressure drops) → datasheet inputs for the equipment (Mechanical), relief loads (Relief & vent) and instruments (Instrumentation & control). Documents cited, not repeated: PFD-100, PVC-PID-001/002, PVC-V1-IN-CE-001, PVC-V1-IN-SRS-001, PVC-V1-SA-HZP-001, the SOP register.

Topic	Continuous unit	Batch reactor R-101
Basis of the balance	one steady state (design case) in kg/h	per batch in t/batch, step by step, plus rates at the governing moments
Streams	flow all the time	most streams exist in one step only (charges, degassing, discharge); utilities follow the heat-release curve
Design point	the design case, ± turndown	the design state (2.0 h, X 0.26) for normal operation; the peak heat release (X 0.72) for cooling; the step rates for charge, degassing and discharge lines
Off-design cases	feed or utility changes at steady state	whole batches run with a different CW temperature or initiator dose: what changes is batch time, peak duty and which valve saturates
Closure	in = out	in = out + accumulation over a cycle that starts and ends with an empty reactor
Utilities	average consumption	per-batch quantity (m³, t, kWh) and the peak rate, then the four reactors staggered for the site

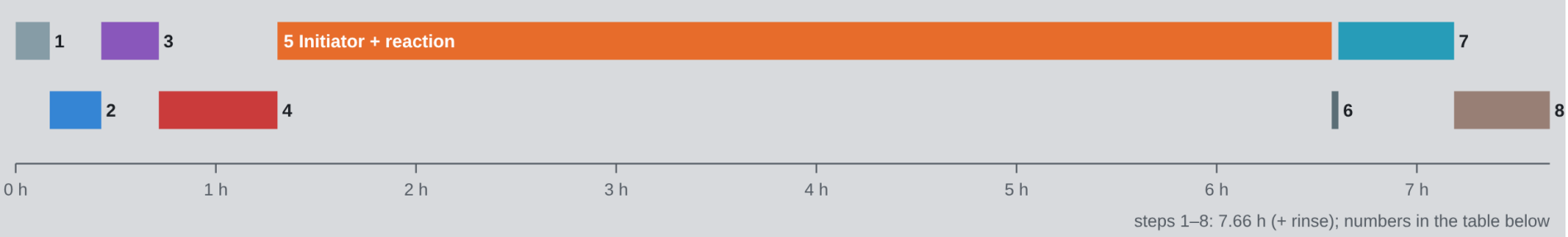
2 Heat and material balance

2.1 Stream table

Numbering of PFD-100 (process streams 1–30) and of the Process tab (block-diagram streams 1–16). "—" = not drawn on PFD-100. Per batch in t unless stated; rates at the design state (2.0 h, X 0.26) and at the peak heat release of the reference batch (X 0.72).

PFD-100	Process tab	Stream	From → to	Phase	Per batch, t	Design 2.0 h	Peak X 0.72	Rate unit	When	T, °C	P, bar g	Composition
6	1	**Demineralised water + PVA**	header → FV-101 / XV-101 → R-101	L	60.0	240.0	—	m³/h	charge (15.4 min)	55.0	5.00	water, PVA ≈ 0.1 % on VCM
3	2	**VCM charge**	TK-100 → P-103 → FV-102 / XV-102 → R-101	L	51.0	180.0	—	t/h	charge (17.3 min)	25.0	11.99	vinyl chloride
8	3	**Initiator**	D-104 → P-104 → XV-103 → R-101	L	—	—	—	—	60 s at the start of the reaction	20.0	—	peroxides A (fast) : B (slow) = 1 : 0.5 charge units
—	4	**Shrinkage water injection**	header → FV-103 → R-101	L	22.0	3.24	5.48	m³/h	reaction (LIC-100 → FIC-103)	30.0	14.99	water
10	5	**VCM vapour to E-101**	R-101 → E-101	V	66.2	12.9	22.8	t/h	reaction	57.0	8.62	VCM + inerts 2.4 mol %
12	6	**Reflux condensate**	E-101 → R-101	L	66.2	12.9	22.8	t/h	reaction	54.0	8.62	liquid VCM
13	7	**E-101 inert vent**	E-101 → HV-135 → VCM recovery	V	0.00	0.00	0.00	kmol/h	on demand (HIC-135 closed in the reference batch)	54.0	8.62	N_2/air + VCM
14 → 15	8	**Degassed VCM**	R-101 → XV-105 / PV-105 → VCM recovery (D-106)	V	6.25	—	18.0	t/h	degassing (34.7 min)	57.0	6.81	VCM (unreacted)
23	9	**PVC slurry**	R-101 → XV-106 → D-102	S	126.5	—	301.6	t/h	discharge (28.7 min)	56.9	1.50	PVC 43.1 t (34 wt %), VCM 1.48 t
22	10	**Jacket loop**	P-101A/B → SM-101 → R-101 jacket	L	—	472.7	472.7	m³/h	whole cycle	38.2	—	tempered water
—	11	**CW injected into the loop**	CWS → TV-110B → loop (spill to CWR)	L	984.7	209.0	215.3	m³/h	reaction	32.0	—	water
—	12	**LP steam**	LS → TV-110A → SM-101	V	1.81	0.00	7.20	t/h	heat-up (peak) and early reaction	148.0	3.50	steam
—	13	**Baffle cooling water**	CWS → FV-120 → baffles → CWR	L	421.3	80.0	80.0	m³/h	reaction	32.0	—	water
—	14	**E-101 cooling water**	CWS → FV-130 → E-101 → CWR	L	212.9	36.3	91.3	m³/h	reaction	32.0	—	water
—	15	**N_2 pad for discharge**	N_2 header → HV-141 → R-101	V	11.2 kmol	—	—	kmol	discharge	20.0	1.50	nitrogen
9	16	**Short-stop**	D-105 → XV-104 → R-101	L	—	—	—	—	end point / SIS	20.0	—	radical scavenger, 10 units
18 → 19	—	**Evacuated air**	R-101 → XV-109 → VP-101 → vent gas treatment	V	0.16 t	—	—	—	evacuation (10.2 min)	30.0	vacuum	air 5.41 kmol

2.2 Per-batch balance by step (reference batch)



Step	Start, min	Duration, min	In, t	Out, t	Rate or quantity	Ends when	Source
1 Evacuation	0.0	10.2	—	air 0.157	5.41 kmol air to vent	VP-101 pump-down to 0.05 bar a	charge run (XV-109)
2 Water + PVA charge	10.2	15.4	water 60.00	—	240 m³/h (FIC-101)	FQI-101 preset 60 t	charge run; totaliser 60.01 t
3 VCM charge	25.6	17.3	vcm 51.00	—	180 t/h (FIC-102)	FQI-102 preset 51 t	charge run; totaliser 51.00 t
4 Heat-up on LP steam	42.9	35.6	—	—	steam 1.75 t (peak 7.2 t/h)	46.2 → 56.8 °C	charge run (TIC-110 AUTO)
5 Initiator + reaction	78.5	316.0	water 22.02	—	injection 22.02 t; VCM → PVC 43.24 t; reflux 66 t	PD-100 end point (1.8 bar drop)	pvBatch({}) reference batch
6 Short-stop	394.5	2.0	water 0.003	—	D-105 → XV-104 (10 units)	2 min mixing	continued run
7 Degassing to recovery	396.5	34.7	—	vcm 6.25	peak 18.0 t/h (HIC-105 100 %)	6.81 → 1.50 bar g	continued run (XV-105)
8 Discharge to D-102	431.2	28.7	—	slurry 126.50, heel 0.284	peak 302 t/h	N_2 pad 1.5 bar g, until empty	continued run (XV-106)

Closure over the cycle (water + VCM + PVC): in 133.026 t = out 133.027 t (degassed VCM 6.246 t + slurry 126.497 t + heel 0.284 t) + accumulation 0 (empty → empty). Residual −0.4 kg = −0.0003 % ✓. The heel is what the model sets to zero when less than 0.3 t is left; it goes to D-102 with the rinse water (assumed). Air (5.41 kmol) and N_2 (11.2 kmol) are below 0.01 % of the mass and are not in the closure.

Component	In, t	Made (+) / used (−), t	Out, t	In + made − out	Notes
Water	82.026	0.000	82.074	−0.047 (−0.058 %)	charged 60 + injected 22.03
VCM	51.000	−43.235	7.735	0.030 (0.059 %)	to recovery 6.25, in the slurry 1.48
PVC	0.000	+43.235	43.218	0.017 (0.039 %)	made in the reaction step

2.3 Design case and off-design batches

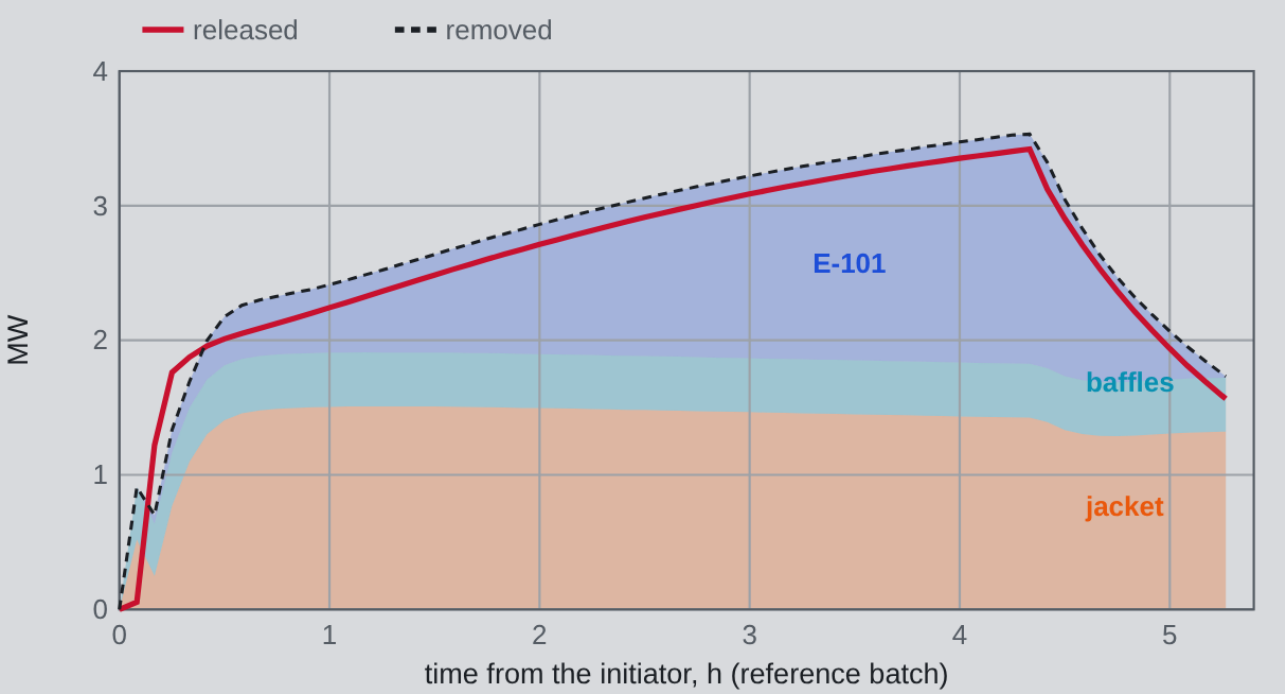
Off-design cases are corners of the Operating windows grid run with the SIS in service: A = CW supply 38 °C (warm day), B = initiator dose × 1.3. Same recipe, same controllers, no operator action.

Quantity	Unit	Reference (design case)	A: CW 38 °C	B: dose × 1.3
CW supply	°C	32	38	32
Initiator dose	× recipe	1.0	1.0	1.3
Reaction time to the PD-100 end point	h	5.27	5.27	4.21
Conversion at the end point	%	84.8	84.7	84.4
PVC per batch	t	43.2	43.2	43.1
Peak heat release	MW	3.43	3.46	4.59
… at conversion	%	72	72	72
Largest heat removal	MW	3.54	3.51	4.57
Largest E-101 duty	MW	1.71	1.99	2.57
Peak FIC-130 (E-101 cooling water)	m³/h	91	540	460
Highest TV-110B (jacket CW valve)	%	54.7	100.0	100.0
Largest VCM boil-up to E-101	t/h	22.8	26.5	34.3
Largest total CW flow	m³/h	387	1016	934
Largest injection (FIC-103)	m³/h	5.50	5.54	7.23
Highest TI-100 after 15 min	°C	57.33	57.10	57.22
Highest PI-100	bar g	8.69	8.64	8.66
K-value of the batch	—	66.99	67.01	67.01
SIS trip	—	none	none	none

What changes. A warm CW day (A) does not change the chemistry: the batch takes 5.27 h as before, but every sink loses driving force, so TV-110B goes to 100 % and VPC-130 pushes E-101 to 540 m³/h — FV-130 wide open. The temperature still holds (57.10 °C) with nothing left in hand. A strong initiator charge (B) shortens the batch to 4.21 h (more PVC per reactor-hour) and raises the peak to 4.59 MW (+34 %): TV-110B 100 %, FIC-130 460 m³/h, injection 7.2 m³/h. Both batches end at the same conversion and K-value; the HMB per batch hardly moves, the instantaneous duties do. That is why a batch HMB needs off-design cases on the peak, not on the totals.

3 Heat balance of the batch and utilities

Reaction step (reference batch)	MJ	% of Q_{rx}
Heat of polymerisation (71 kJ/mol)	49115	100.0
Agitator work (95 % to the slurry)	5400	11.0
Injection water heated 30 °C → T	−2485	−5.1
Jacket	−26274	−53.5
Baffles	−7599	−15.5
Reflux condenser E-101	−17845	−36.3
VCM flashed to vents	0	0.0
Losses to ambient	−307	−0.6
Stored in the contents (Σ C·dT)	−5	0.0
Residual (in − out − stored)	**−0.1**	**−0.0003**



Heat released and removed over the reference batch (jacket, baffles, E-101 stacked).

3.1 Duties at the governing moments

Duty, MW	Design state (2.0 h, X 0.26)	Peak (reference, X 0.72)	A: CW 38 °C (largest)	B: dose × 1.3 (largest)	Installed, peak state, 57 °C, wide open
Heat of polymerisation released	2.71	3.43	3.46	4.59	—
Jacket	1.50	1.43	1.22	1.60	1.60
Baffles	0.40	0.40	—	—	0.42
Reflux condenser E-101	0.97	1.71	1.99	2.57	2.58
Total removed	2.87	3.54	3.51	4.57	**4.60**
Agitator work into the slurry	0.28	0.31	—	—	—
TV-110B / FIC-130	53 % / 36	54 % / 91	100 % / 540	100 % / 460	100 % / 540

3.2 Utility summary

Utility	Per batch	Peak rate (reference)	Notes	Specific
Cooling water (jacket injection + baffles + E-101)	1708 m³	387 m³/h	A 1016 / B 934 m³/h	39.5 m³/t PVC
LP steam (3.5 bar g)	3.17 t	7.2 t/h	heat-up 1.75 t, reaction 0.06 t, short-stop to discharge 1.36 t	73 kg/t PVC
Electric power (agitator, P-101, P-103)	2777 kWh	≈ 414 kW	agitator 2086, P-101 674, P-103 16 kWh; peak = agitator at the peak heat release + one P-101 (econ())	64 kWh/t PVC
Nitrogen (discharge pad)	11.2 kmol	—	HV-141 to 1.5 bar g while the slurry leaves	7.2 kg/t PVC
Four reactors, staggered (average)	837 m³/h CW	1.55 t/h steam	cycle 8.16 h = steps 1–8 + 0.5 h rinse (assumed, Transition tab)	169 kt/y PVC

4 Line sizing

Each line is sized for its design flow (a charge set point, a valve wide open, or the peak of the worst batch) against a typical criterion for its service. Schedule 40 commercial steel, roughness 0.045 mm; Darcy friction factor from Swamee–Jain; densities from the model, viscosities typical. Gas densities ideal (Z = 1, conservative for ρv²).

Service	Criterion (typical, assumed for this unit)
Liquid, pump or header discharge	1–3 m/s, Δp ≤ 90 kPa/100 m
Pump suction	0.3–1.5 m/s, Δp ≤ 25 kPa/100 m
Cooling water	1–3 m/s, Δp ≤ 50 kPa/100 m
Gravity return (self-venting)	Froude v/√(gD) ≤ 0.3
Vapour or gas	ρv² ≤ 6 000 Pa
Low-Δp vapour (to a condenser or a low-pressure header)	ρv² ≤ 6 000 Pa, Δp ≤ 5 kPa/100 m
LP steam	v ≤ 35 m/s
Slurry	v ≥ Durand deposit velocity (F_L 1.0, s 1.4) and ≤ 3.5 m/s
Relief discharge	sized in PVC-V1-PE-RLB-001 (Relief & vent tab)

4.1 Line list

Line	Service	PFD	Design flow (basis)	ρ, kg/m³	μ, mPa s	Size	ID, mm	v, m/s	Re	f (Darcy)	Δp, kPa/100 m	ρv² Pa / Fr / v_D	Criterion	OK
L-101	Demineralised water + PVA charge	6	240 m³/h (FIC-101 set point)	988.3	0.500	8"	202.7	2.07	8.28×10^5	0.0152	15.8	—	Liquid, pump or header discharge	✓
L-102	VCM suction TK-100 → P-103	2	180 t/h (FIC-102 set point)	902.0	0.180	10"	254.5	1.09	1.39×10^6	0.0142	3.0	—	Pump suction	✓
L-103	VCM charge P-103 → R-101	3	180 t/h (FIC-102 set point)	902.0	0.180	6"	154.1	2.97	2.30×10^6	0.0153	39.5	—	Liquid, pump or header discharge	✓
L-104	Shrinkage water injection	—	10 m³/h (FIC-103 range; peak used 7.2)	997.0	0.800	2"	52.5	1.28	8.40×10^4	0.0222	34.8	—	Liquid, pump or header discharge	✓
L-105	Jacket loop P-101A/B discharge	22	520 m³/h (both pumps, K.Fc × 1.1; one pump 473)	990.0	0.650	10"	254.5	2.84	1.10×10^6	0.0144	22.6	—	Liquid, pump or header discharge	✓
L-106	CW to the jacket loop	—	396 m³/h (TV-110B wide open, K.Fcw110)	996.5	0.770	10"	254.5	2.17	7.15×10^5	0.0148	13.7	—	Cooling water	✓
L-107	CW to the baffles	—	198 m³/h (FV-120 wide open, K.Fcw120)	996.5	0.770	8"	202.7	1.71	4.49×10^5	0.0159	11.4	—	Cooling water	✓
L-108	CW to E-101	—	540 m³/h (FV-130 wide open, K.Fcw130)	996.5	0.770	12"	303.2	2.08	8.18×10^5	0.0144	10.3	—	Cooling water	✓
L-109	VCM vapour R-101 → E-101	10	37.7 t/h = 1.1 × 34.3 t/h (largest boil-up of the three batches)	21.4	0.011	12"	303.2	6.77	4.00×10^6	0.0133	2.2	982	Low-Δp vapour (to a condenser or a low-pressure header)	✓
L-110	Reflux condensate E-101 → R-101	12	37.7 t/h liquid VCM	850.5	0.170	8"	202.7	0.38	3.87×10^5	0.0161	0.5	Fr 0.27	Gravity return (self-venting)	✓
L-111	E-101 inert vent (upstream of HV-135)	13	25.9 kmol/h = 1.51 t/h (HV-135 wide open at 8.62 bar g)	20.6	0.012	2"	52.5	9.38	8.45×10^5	0.0194	33.5	1812	Vapour or gas	✓
L-112a	Degassing R-101 → PV-105	14	18.0 t/h at 6.44 bar g (start of degassing, HIC-105 100 %)	17.0	0.011	6"	154.1	15.75	3.75×10^6	0.0151	20.7	4216	Vapour or gas	✓

Line	Service	PFD	Design flow (basis)	ρ, kg/m³	μ, mPa s	Size	ID, mm	v, m/s	Re	f (Darcy)	Δp, kPa/100 m	ρv² Pa / Fr / v_D	Criterion	OK
L-112b	Degassing PV-105 → recovery header	15	18.0 t/h at 1.25 bar a (header)	2.9	0.011	12"	303.2	24.23	1.90×10^{6}	0.0137	3.8	1675	Low-Δp vapour (to a condenser or a low-pressure header)	✓
L-113	Slurry discharge R-101 → D-102	23	302 t/h (start of discharge), ρ 1096 kg/m³	1096.0	3.000	8"	202.7	2.37	1.75×10^{5}	0.0176	26.7	v_D 1.26	Slurry	✓
L-114	LP steam to TV-110A / SM-101	—	7.2 t/h (TV-110A wide open, K.Fst), 3.5 bar g saturated	2.3	0.014	8"	202.7	26.68	8.97×10^{5}	0.0151	6.2	1653	LP steam	✓
L-115	PSV-100 discharge	20	relief load from PVC-V1-PE-RLB-001	—	—	—	—	—	—	—	—	—	Relief discharge	ref.

4.2 Worked example: L-103 VCM charge

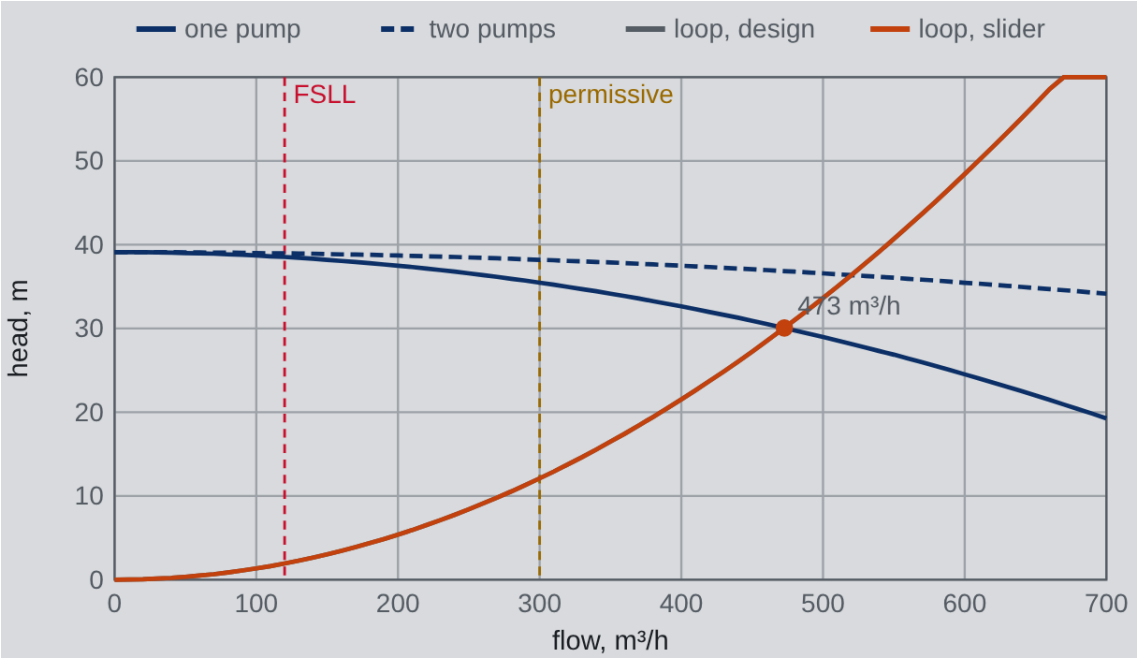
Step	How	Result
Mass flow	FIC-102 set point	180 t/h = 50.00 kg/s
Density	model $\rho_{VCM} = 947 - 1.8\cdot T$ at 25 °C	902 kg/m³
Volume flow	$Q = \dot{m}/\rho$	0.0554 m³/s = 199.6 m³/h
Pipe	6" sch 40, ID 154.05 mm	$A = \pi D^2/4 = 0.01864$ m²
Velocity	$v = Q/A$	**2.97 m/s**
Reynolds number	$Re = \rho v D/\mu$, μ = 0.18 mPa s (typical, liquid VCM at 25 °C)	2.30×10^{6} (turbulent)
Relative roughness	ε/D = 0.045 mm / D	2.92×10^{-4}
Friction factor	Swamee–Jain $f = 0.25 / [\log_{10}(\varepsilon/3.7D + 5.74/Re^{0.9})]^2$	0.0153
Pressure drop	$\Delta p/L = f\,(1/D)\,\rho v^2/2$	**39.5 kPa/100 m**
Check	1–3 m/s, Δp ≤ 90 kPa/100 m	✓ within 1–3 m/s, Δp ≤ 90 kPa/100 m
Why not 4" or 8"	same flow	4": v 6.75 m/s, Δp 331 kPa/100 m · 8": v 1.72 m/s, Δp 9.6 kPa/100 m

5 Pump hydraulics

5.1 P-101A/B jacket circulation

The model holds the loop at K.Fc = 130 kg/s (473 m³/h on FI-110) with one pump and 143 kg/s with two; it has no loop hydraulics. The engineering view: a closed loop has no static head, so the system curve is H = kQ² through the loop Δp at the rated flow (element Δp assumed, piping from L-105). Two identical pumps in parallel give $Q_2/Q_1 = \sqrt{[(a + k)/(a/4 + k)]}$; the model's factor 1.1 fixes a = 0.301 k, i.e. a shut-off head 1.30 × the rated head — a normal, fairly flat centrifugal curve. Raise the jacket Δp (plugged nozzles, scale) to see the flow fall toward the 300 m³/h permissive and the 120 m³/h FSLL-110 trip.

P-101A/B loop (design settings)	Value
Jacket (agitation nozzles, spiral baffle; assumed at 473 m³/h)	2.20 bar
SM-101 steam mixer (assumed)	0.30 bar
Strainer, check valve, isolation valves (assumed)	0.25 bar
Piping L-105 10" sch 40, 90 m equivalent (v 2.58 m/s, 18.8 kPa/100 m)	0.169 bar
Loop Δp at the rated flow → head	**2.92 bar → 30.1 m**
Pump curve $H = H_s - aQ^2$ through the rated point; shut-off head	39.1 m = 1.30 × rated head
Absorbed power at the rated point (η 0.8 assumed)	47.9 kW (econ() charges 90 kW per running pump)
Operating point with the slider settings	473 m³/h at 30.1 m
Two pumps ÷ one pump (same system)	1.100 (model: K.Fc × 1.1)



5.2 P-103 VCM charge pump

TK-100 stores VCM as a liquid at its own vapour pressure (3.98 bar a at 25 °C), so the tank pressure gives no NPSH: only the liquid height above the pump does, minus the suction losses. Any heat picked up in the suction line raises the vapour pressure at the impeller and eats the margin. The discharge side is the model's: P-103 delivers K.Pvcm = 13 bar a and FV-102 with the line (lumped K.Cv102 = 34) takes the difference to the reactor pressure.

P-103 (design settings)	Value
TK-100 at 25 °C, saturated: pressure = VCM vapour pressure (model, DIPPR)	3.980 bar a
Static head, TK-100 low level above the pump (assumed)	4.50 m
Suction losses: L-102 10" 30 m eq. + strainer 0.05 bar	0.67 m
Vapour pressure rise from suction heat gain 0.0 °C	0.00 m
NPSH available = $z - h_f - \Delta P_v/\rho g$	**3.83 m**
NPSH required (assumed, 200 m³/h end-suction pump) + margin 1 m (typical)	3.50 m ✓
Suction pressure at the pump	4.32 bar a
Discharge pressure (model: K.Pvcm, delivered by P-103)	13.00 bar a = 12.0 bar g
Differential pressure → head	8.68 bar → 98 m
Absorbed power at 180 t/h (η 0.7 assumed)	**68.7 kW** vs 55 kW in econ() ✗

FV-102 during the VCM charge (charge run): the reactor pressure jumps to 7.38 bar g as the first VCM flashes, then falls as the VCM dissolves and cools; FV-102 opens at most 67.6 % ✓ (≤ 80 % at the design flow, typical). Δp left for FV-102 at the highest reactor pressure = 13.00 − 8.39 (reactor) − 0.88 (static, 10 m assumed) − 0.32 (L-103, 80 m eq.) = **3.41 bar**.

Time into the charge, min	FV-102, %	PI-100, bar g	FI-102, t/h
0.3	37.1	1.33	149
1.4	66.8	7.34	176
2.4	67.6	7.24	180
5.3	65.9	6.99	180
8.3	64.6	6.79	180
11.3	63.7	6.64	180
14.3	63.1	6.55	180
16.4	62.9	6.52	180

6 Design basis and margins

A batch unit is sized at different moments for different equipment: the cooling at the peak heat release, the charge pumps at the charge time the cycle allows, the degassing line at the start of degassing, the discharge line at the start of discharge.

Item	Sized by (moment / case)	Normal (reference batch)	Installed / design	Margin
Heat removal (jacket + baffles + E-101)	peak heat release of the reference batch (X 0.72)	3.43 MW	4.60 MW at 57 °C, CW 32 °C; 3.47 MW at CW 38 °C	**1.34** (typical ≥ 1.25); 1.01 at 38 °C; case B peak 4.59 MW
E-101 cooling water FV-130	peak of the worst batch (A: CW 38 °C)	91 m³/h (reference)	540 m³/h (K.Fcw130); FIC-130 range 600	A reaches 540 m³/h: no margin left on a warm day
Jacket CW TV-110B	VPC-130 keeps it near 50 % by design	54.7 %	396 m³/h wide open	headroom 45 % in the reference; 100 % (A), 100 % (B)
Jacket circulation P-101A/B	U of the jacket, FSLL-110	473 m³/h one pump	trip 120 m³/h, permissive 300 m³/h, two pumps 520	3.9 × the trip flow
Water charge (header, FV-101)	charge time in the cycle	60 t in 15.4 min	240 m³/h; FIC-101 range 300	range 1.25 × set point
VCM charge P-103, FV-102	charge time in the cycle	51 t in 17.3 min	180 t/h; FIC-102 range 250; FV-102 at most 68 %	valve 32 % travel spare
Shrinkage injection FV-103	largest injection of the three batches	5.50 m³/h	FIC-103 range 10 m³/h	B needs 7.23 m³/h → 1.38 ×
Heat-up steam TV-110A	heat-up time	35.6 min, 1.75 t	7.2 t/h wide open (K.Fst)	TV-110A reaches 100 %: the steam rate sets the heat-up time
Degassing L-112, PV-105	start of degassing (highest flow)	18.0 t/h	to 1.25 bar a header; 34.7 min to 1.5 bar g	four reactors must not degas at once (Transition tab)
Discharge L-113, XV-106	start of discharge (highest flow)	302 t/h	28.7 min, N_2 pad 1.5 bar g	flow falls with the head to 221 t/h at the end: v 1.74 m/s vs deposit 1.26 m/s ✓
Reactor pressure	highest PI-100 of the three batches	8.69 bar g	PSHH-101 12.5, PSV-100 15, design 15.2 bar g	3.81 bar to the trip
Level	highest level, reference batch	83.9 %	LSHH-100 94 %	10.1 % of height

6.1 Worked calculation: heat-removal capacity at the peak

Step	Calculation
Peak state (reference batch)	t = 4.37 h, X = 0.720, T = 57.01 °C, Q_{rx} = 3428 kW
Jacket UA from the state	UA = $Q_j/(T - T_{jm})$ = 1427/(57.01 − 39.06) = **79.5 kW/K**
TV-110B wide open	F_{cw} = 110 kg/s, e = $F_{cw}/(F_c + F_{cw})$ = 0.458
Jacket duty	Q = UA(T − T_{cw})/[1 + UA/($F_c c_p$)(1/e − ½)] = **1596 kW** (now 1427 kW)
Baffles, FV-120 wide open	UA_b = 17.6 kW/K, C_b = 230 kW/K: Q = $C_b(1 - e^{-UA/C})(T - T_{cw})$ = **424 kW** (now 401)
E-101, FV-130 wide open	T_{cond} = 56.09 °C, UA_c = 130·(1 − b 0.098) = 117.3 kW/K, C_c = 627 kW/K: **2577 kW** (now 1709)
Total ÷ peak release	4596 / 3428 = **1.34**; at CW 38 °C 3470 kW → 1.01
Same equations at the present handle positions	1429 + 401 + 1709 kW vs the model's 1427 + 401 + 1709 kW (validates the calculation)

7 Checks against the model and the other documents

67 ✓ and 3 ✗ of 70 checks, computed in the tab when the document was generated (including a live re-run of the reference batch). A ✗ is a finding, listed as an engineering query below.

#	Check	Compared with	This document	Other value	Result
1	Per-batch mass balance closes (in − out − accumulation)	this HMB, steps 1–8	133.026 − 133.027 − 0 t	|residual| 0.0003 % < 0.1 %	✓
2	Component balance: Water	this HMB	−0.047 t	0.058 % < 0.1 %	✓
3	Component balance: VCM	this HMB	0.030 t	0.059 % < 0.1 %	✓
4	Component balance: PVC	this HMB	0.017 t	0.039 % < 0.1 %	✓
5	Heat balance of the reaction step closes	integrated model terms	−0.1 MJ	0.0003 % of the heat released < 0.1 %	✓
6	HMB design-state values = snapDesign of this page	model (snapDesign, 2.0 h)	12 quantities (T, P, duties, flows)	largest difference 0.158 % (Fcw3); tolerance 0.1 %	✗
7	HMB shrinkage injection = snapDesign of this page (the page warms its design state up from a noisy first measurement: FIC-103 (slave of LIC-100) moves by up to ≈ 0.6 % from one page load to the next)	model (snapDesign, 2.0 h)	3.242 m³/h	3.217 m³/h (tolerance 2 %)	✓
8	Stream 4 (Shrinkage water) design rate = Process tab	Process tab streamRows()	3.24 m³/h	3.22 m³/h (tolerance 2 %)	✓
9	Stream 5 (VCM vapour) design rate = Process tab	Process tab streamRows()	12.95 t/h	12.96 t/h (tolerance 0.5 %)	✓
10	Stream 10 (Jacket loop) design rate = Process tab	Process tab streamRows()	472.73 m³/h	472.73 m³/h (tolerance 0.5 %)	✓
11	Stream 11 (CW injected) design rate = Process tab	Process tab streamRows()	208.96 m³/h	208.91 m³/h (tolerance 0.5 %)	✓
12	Stream 13 (Baffle CW) design rate = Process tab	Process tab streamRows()	80.00 m³/h	80.00 m³/h (tolerance 0.5 %)	✓
13	Stream 14 (E-101 CW) design rate = Process tab	Process tab streamRows()	36.34 m³/h	36.39 m³/h (tolerance 0.5 %)	✓
14	PFD-100 stream 3 flag: VCM charge, t/h	PFD-100 (PF_SPEC)	180.00	180.00	✓
15	PFD-100 stream 3 flag: P-103 discharge, bar g	PFD-100 (PF_SPEC)	11.99	12.00	✓
16	PFD-100 stream 6 flag: water charge, m³/h	PFD-100 (PF_SPEC)	240.00	240.00	✓
17	PFD-100 stream 6 flag: water temperature, °C	PFD-100 (PF_SPEC)	55.00	55.00	✓
18	PFD-100 stream 10 flag: VCM vapour to E-101, t/h	PFD-100 (PF_SPEC)	12.95	12.90	✓
19	PFD-100 stream 10 flag: reactor pressure, bar g	PFD-100 (PF_SPEC)	8.62	8.60	✓
20	PFD-100 stream 22 flag: jacket circulation, t/h	PFD-100 (PF_SPEC)	468.00	468.00	✓
21	PFD-100 stream 22 flag: jacket inlet, °C	PFD-100 (PF_SPEC)	38.17	38.20	✓
22	Peak heat release = reference batch (Design tab)	Design tab DS_REF	3429 kW at X 0.720	3429 kW at X 0.720	✓
23	Reaction time = reference batch (Design tab)	Design tab DS_REF	5.267 h	5.267 h	✓
24	Embedded runs = live re-run of pvBatch({}) in this page	model, run now (0.8 s)	5.267 h, 3429 kW, 91.4 m³/h	5.267 h, 3429 kW, 91.4 m³/h	✓
25	Peak, injection and degassing basis vs the Process tab reference batch	Process tab refBatch()	peak 3429 kW, injection 22.02 t, PVC 43.24 t	peak 3429 kW, injection 22.02 t, PVC 43.24 t	✓
26	Water charge time = Transition tab (cold batch)	Transition tab TR_D	15.4 min	15.5 min	✓
27	VCM charge time = Transition tab (cold batch)	Transition tab TR_D	17.3 min	17.2 min	✓
28	Degassing time to 1.5 bar g = Transition tab (cold batch)	Transition tab TR_D	34.7 min	34.5 min	✓
29	Discharge time = Transition tab (cold batch)	Transition tab TR_D	28.7 min	28.8 min	✓
30	Degassed VCM = Transition tab	Transition tab TR_D	6.25 t	6.23 t	✓
31	Slurry to D-102 = Transition tab	Transition tab TR_D	126.5 t	126.7 t	✓
32	Charge times = preset ÷ set point	FQI presets, FIC set points	15.4 / 17.3 min	15.2 / 17.0 min (+ ramp-up of FIC-101/102)	✓

#	Check	Compared with	This document	Other value	Result
33	Off-design A = Operating windows grid	Operating windows OW_G (SIS off)	5.27 h, 3.46 MW, 540 m³/h	5.27 h, 3.46 MW, 540 m³/h	✓
34	Off-design B = Operating windows grid	Operating windows OW_G (SIS off)	4.21 h, 4.59 MW, 460 m³/h	4.21 h, 4.59 MW, 460 m³/h	✓
35	L-101 Demineralised water + PVA charge: within its criterion	Liquid, pump or header discharge	8" · v 2.07 m/s · Δp 15.8 kPa/100 m	1–3 m/s, Δp ≤ 90 kPa/100 m	✓
36	L-102 VCM suction TK-100 → P-103: within its criterion	Pump suction	10" · v 1.09 m/s · Δp 3.0 kPa/100 m	0.3–1.5 m/s, Δp ≤ 25 kPa/100 m	✓
37	L-103 VCM charge P-103 → R-101: within its criterion	Liquid, pump or header discharge	6" · v 2.97 m/s · Δp 39.5 kPa/100 m	1–3 m/s, Δp ≤ 90 kPa/100 m	✓
38	L-104 Shrinkage water injection: within its criterion	Liquid, pump or header discharge	2" · v 1.28 m/s · Δp 34.8 kPa/100 m	1–3 m/s, Δp ≤ 90 kPa/100 m	✓
39	L-105 Jacket loop P-101A/B discharge: within its criterion	Liquid, pump or header discharge	10" · v 2.84 m/s · Δp 22.6 kPa/100 m	1–3 m/s, Δp ≤ 90 kPa/100 m	✓
40	L-106 CW to the jacket loop: within its criterion	Cooling water	10" · v 2.17 m/s · Δp 13.7 kPa/100 m	1–3 m/s, Δp ≤ 50 kPa/100 m	✓
41	L-107 CW to the baffles: within its criterion	Cooling water	8" · v 1.71 m/s · Δp 11.4 kPa/100 m	1–3 m/s, Δp ≤ 50 kPa/100 m	✓
42	L-108 CW to E-101: within its criterion	Cooling water	12" · v 2.08 m/s · Δp 10.3 kPa/100 m	1–3 m/s, Δp ≤ 50 kPa/100 m	✓
43	L-109 VCM vapour R-101 → E-101: within its criterion	Low-Δp vapour (to a condenser or a low-pressure header)	12" · v 6.77 m/s · Δp 2.2 kPa/100 m	ρv² ≤ 6 000 Pa, Δp ≤ 5 kPa/100 m	✓
44	L-110 Reflux condensate E-101 → R-101: within its criterion	Gravity return (self-venting)	8" · v 0.38 m/s · Δp 0.5 kPa/100 m	Froude v/√(gD) ≤ 0.3	✓
45	L-111 E-101 inert vent (upstream of HV-135): within its criterion	Vapour or gas	2" · v 9.38 m/s · Δp 33.5 kPa/100 m	ρv² ≤ 6 000 Pa	✓
46	L-112a Degassing R-101 → PV-105: within its criterion	Vapour or gas	6" · v 15.75 m/s · Δp 20.7 kPa/100 m	ρv² ≤ 6 000 Pa	✓
47	L-112b Degassing PV-105 → recovery header: within its criterion	Low-Δp vapour (to a condenser or a low-pressure header)	12" · v 24.23 m/s · Δp 3.8 kPa/100 m	ρv² ≤ 6 000 Pa, Δp ≤ 5 kPa/100 m	✓
48	L-113 Slurry discharge R-101 → D-102: within its criterion	Slurry	8" · v 2.37 m/s · Δp 26.7 kPa/100 m	v ≥ Durand deposit velocity (F_L 1.0, s 1.4) and ≤ 3.5 m/s	✓
49	L-114 LP steam to TV-110A / SM-101: within its criterion	LP steam	8" · v 26.68 m/s · Δp 6.2 kPa/100 m	v ≤ 35 m/s	✓
50	FIC-101 range covers the design flow	makeCtrls() range	240.0 m³/h	0–300 m³/h	✓
51	FIC-102 range covers the design flow	makeCtrls() range	180.0 t/h	0–250 t/h	✓
52	FIC-103 range covers the design flow	makeCtrls() range	7.2 m³/h	0–10 m³/h	✓
53	FIC-120 range covers the design flow	makeCtrls() range	198.7 m³/h	0–250 m³/h	✓
54	FIC-130 range covers the design flow	makeCtrls() range	540.0 m³/h	0–600 m³/h	✓
55	FI-110 range covers two pumps	IND (DCS faceplate)	520 m³/h	0–600 m³/h	✓
56	QI-100 range covers the largest heat removal	IND (DCS faceplate)	4.57 MW	0–5 MW	✓
57	P-101 flow above FSLL-110 and the 300 m³/h permissive	K.FII, purgePermissives()	473 m³/h	> 120 and ≥ 300 m³/h	✓
58	P-101 absorbed power ≤ the power the model charges (econ() is ≈ 2 × the hydraulic estimate: see EQ)	econ(): 90 kW per running pump	47.9 kW	90 kW	✓
59	P-103 absorbed power ≤ the power the model charges (13 bar a delivery from saturated storage needs ≈ 70 kW: see EQ)	econ(): 55 kW for P-103	68.7 kW	55 kW	✗
60	P-103 NPSH available ≥ required + 1 m	assumed elevation and NPSHr	3.83 m	3.50 m	✓
61	FV-102 at most 80 % open at the design flow	charge run (model Cv102)	67.6 %	≤ 80 %	✓
62	Capacity equations reproduce the model duties at the peak	model state at the peak	3539 kW	3537 kW	✓
63	Installed heat removal ≥ 1.25 × peak release (CW 32 °C)	Operating windows rule (typical)	1.34	≥ 1.25	✓
64	Installed heat removal ≥ peak release at CW 38 °C (no margin left on a warm day: see EQ)	off-design A	1.01	≥ 1.00	✓
65	Highest pressure of the three batches below PSHH-101	K.Phh	8.69 bar g	12.5 bar g	✓
66	Highest level below LSHH-100	K.Lhh	83.9 %	94 %	✓

#	Check	Compared with	This document	Other value	Result
67	PSV-100 discharge destination = relief load basis	PVC-V1-PE-RLB-001	L-115 → D-103	D-103 emergency vent tank (closed system)	✓
68	R-101 design pressure (equipment list) above the HMB maximum	PVC-V1-ME-EQL-001	8.69 bar g	15.2 bar g	✓
69	P-103 motor (equipment list) ≥ absorbed power	PVC-V1-ME-EQL-001	68.7 kW	75 kW	✓
70	P-103 absorbed power = electrical load list	PVC-V1-EE-LL-001	68.7 kW	55.0 kW	✗

7.1 Engineering queries

#	Subject	Finding	Raised against	Proposed
EQ-EP-01	**P-103 power**	econ() charges 55 kW for the VCM charge pump. The model's own numbers (180 t/h delivered at K.Pvcm = 13 bar a from TK-100 stored saturated at 3.98 bar a, 25 °C) need about 8.7 bar differential, 98 m head and ≈ 69 kW absorbed at an assumed efficiency of 0.70.	Model econ() / PVC-V1-EE-LL-001 (load list) / PVC-V1-ME-EQL-001	Rate P-103 for ≈ 70 kW absorbed (75 kW motor) or confirm a lower delivery pressure; carry the same figure into the equipment and load lists.
EQ-EP-02	**P-101A/B power**	econ() charges 90 kW per running jacket pump; the loop Δp assumed here (jacket 2.2 bar + mixer + valves + 90 m of 10" pipe ≈ 2.9 bar, 30 m) gives ≈ 48 kW absorbed. Either the jacket Δp is about twice the assumption or 90 kW is a motor rating, not a load.	Model econ() / PVC-V1-ME-EQL-001 (jacket Δp) / PVC-V1-EE-LL-001	Confirm the jacket pressure drop at 470 m³/h on the vessel datasheet; state in the load list whether 90 kW is the rating or the absorbed load.
EQ-EP-03	**Degassed VCM in the Process tab stream table**	Process tab stream 8 'Degassed VCM' gives the VCM left in R-101 at the end point (≈ 7.8 t). Degassing to 1.5 bar g sends 6.25 t to recovery; ≈ 1.5 t stays dissolved in the PVC and the water and leaves with the slurry to D-102 and the stripper (Transition tab: 6.23 t vented, 1.48 t in the slurry).	Process tab (stream table, stream 8); Design tab text '≈ 15 % of the charge goes back to recovery'	Show stream 8 as 6.2 t/batch to recovery and add the VCM in the slurry (1.5 t) to stream 9; quote ≈ 12 % of the charge to recovery.
EQ-EP-04	**Utility streams missing on PFD-100**	The shrinkage-water injection (FV-103), the CW to TV-110B, the baffle CW (FV-120), the E-101 CW (FV-130) and the N_2 pad have no PFD-100 stream numbers (note 3 of the drawing omits baffle CW and the N_2 pad). The HMB carries them with the Process tab numbers only.	PFD-100 rev 0	Number the utility streams on PFD-100 rev 1 (or add a utility flow diagram) so the HMB and the line list can reference one number.
EQ-EP-05	**Cooling margin on a warm day**	Heat-removal capacity at the peak state with every handle wide open is 1.34 × the peak heat release at CW 32 °C but only 1.01 × at 38 °C; off-design A drives FV-130 to 540 m³/h (wide open) and TV-110B to 100 %. The cooling is sized at the 32 °C design CW temperature with no allowance for summer.	Design tab (design basis) / Operating windows (CW range)	Confirm the CW design supply temperature for the site (summer); if 38 °C occurs, either add E-101 area or reduce the initiator dose on warm days (procedure).
EQ-EP-06	**Cycle-time basis**	The plant capacity (200 kt/y) uses a 6.9 h cycle (model card). The scripted cycle of this HMB is 7.7 h for steps 1–8 (Transition tab cold batch 7.75 h) plus the rinse, i.e. about 170 kt/y with four reactors at 8 000 h/y.	Model card / RESUME basis; Transition and Process reliability tabs	Fix one cycle-time basis for all documents (already an open item between Process reliability 6.9 h and Transition 7.75 h).

7.2 Assumptions

• Element pressure drops of the P-101 loop (jacket 2.2 bar, mixer 0.3 bar, valves and strainer 0.25 bar, 90 m equivalent pipe) and pump efficiency 0.80.

• P-103: TK-100 low level 4.5 m above the pump, 30 m equivalent suction line, clean strainer 0.05 bar, NPSHr 2.5 m, margin 1 m, efficiency 0.70; discharge 10 m static, 80 m equivalent line.

• Viscosities: water 0.50–0.80 mPa s, liquid VCM 0.17–0.18 mPa s, vapours 0.011–0.014 mPa s, slurry 3 mPa s (apparent).

• Line criteria as in section 4 (typical practice); 10 % margin on the largest boil-up for the E-101 vapour and reflux lines.

• Cycle for the plant figures = steps 1–8 + 0.5 h rinse (Transition tab assumption); 8 000 h/y.

• Typical practice quoted: line sizing by velocity and Δp per 100 m; NPSH margin per API 610 / pump-standard practice; relief per API 520/521 (in PVC-V1-PE-RLB-001).